

AI Working Group (AI WG) Meeting Notes

September 20, 2024

Agenda

- Status update on working group, collaborative brainstorm for prompt injection related weakness, discussion on possible documentation for this group, and sub-group opportunities.

Action Items

- Reach out to participants to see who would be interested in tackling some of the tasks mentioned and forming sub-working groups.
- Develop mission statement, charter, and guiding principles for the WG.

Summary

- **Meeting Overview:** The AI Working Group convened on September 20, 2024, to discuss AI-related topics, including sub-working groups, new CWE required elements, CWE entry development, and documentation for the CWE AI WG.
- **Formation of Sub-Working Groups:** The group emphasized the need to form sub-working groups and proposed two possible subgroups.
- **Development of Entry:** Much of the time was applied to developing potential consequences, mitigations, detection, applicable platforms, and modes of introduction for a new CWE entry being drafted by the group related to prompt injection.
- **Future Tasks and Submissions:** Future tasks include crafting updated solutions for existing weaknesses relevant to AI, especially for the continued development of a CWE entry for Improper [Missing or Incorrect] Neutralization of Input Used for LLM Prompting, and the finalization of the entry for submission on October 24. Member also proposed developing onboarding resources, a charter, and a discussion of guiding principles for the WG at the next meeting on September 27, 2024.

Meeting Notes

Connor Mullaly facilitated the discussion and went through the meeting agenda and proceeded to highlight the new CWE required elements highlighted in green.

New Entry Development Discussion

Connor then reinitiated the discussion from the previous meeting for a proposed CWE entry: *“Improper [Missing or Incorrect] Neutralization of Input Used for LLM Prompting.”*

Common Consequences Worksheet

Connor presented the common consequences indicated by the submitter of the proposed CWE entry, which included confidentiality, integrity, and access control, for group discussion.

Consequence 1, “Execute Unauthorized Code or Commands” was discussed last week. Consequences 2-4 were submitted on September 19. Member asked whether there was a way of scoping expected ranges of responses from LLMs. Connor responded that it would be appropriate to get more targeted language in the worksheet.

Member noted that the team discussed the approach taken at DEFCON, where the model card was utilized to make small modifications to statements of intent, which was well received by the Allen Institute.

There was a brief exchange between members about whether the model card hacking was cybersecurity related or safety related, but this discussion was tabled.

Connor moved on from the Common Consequences Worksheet and recommended that participants keep discussing consequences in the CDR.

Potential Mitigations Worksheet

Connor moved onto a discussion of potential mitigations, much of which was discussed at the last meeting on September 19.

Member mentioned that one potential mitigation that was missing was guardrails, such as preventing certain actions from taking place. Member mentioned that guardrails should be implemented during the deployment phase of the software development cycle.

A brief exchange followed about the use of the phrase “ensure that the troublesome behavior is not possible” under the description for Mitigation 3, as output validation was not discussed in the context of this entry development.

Member asked for clarification about “misleading or dangerous characters” in Mitigation 1. Connor Mulally agreed that the phrase is limiting, as it may not always include characters. Member suggested that the phrase might need expanding or addition. V added that there may be legal implications in industry for the phrasing. Member suggested the word “subversive,” while Member suggested “adversarial.”

Modes of Introduction

Connor presented the Modes of Introduction Worksheet and pointed out the input validation and whether the removal of mark up tags was relevant.

Member replied that it seemed a little too specific, and depended on the design of the system, and Member agreed.

Member said that the only thing that would work in human language is using semantic comparators against the intent of the publisher, which quickly becomes unenforceable.

Member wondered whether all LLMs were vulnerable to prompt injection.

Member asked whether the idea of query intent was being studied. This research language could be folded into mitigation language.

Member said this was a hard problem, and something that was being studied.

Connor Mulally went over the modes of introduction discussion from the previous week.

Observed Examples Worksheet

The submitter provided three CVE IDs for observed examples, all of which Member identified as good examples for further discussion.

Demonstrative Examples Worksheet

Connor opened the floor to any participant who wanted to submit examples, but there were no suggestions from the group.

Next Steps

Continue to develop and draft remaining content. MITRE to move submission of CWE to “Accepted” and later to “Full Details Accepted.” MITRE to continue coordinating with submitter and AI WG to develop this submission. Discussion to take place in CDR. Next CWE release scheduled for October 24.

Documentation and Governance:

- **CWE AI WG Onboarding Document Development:** Kate Farris to lead the creation of essential documents including a mission statement and resources for new and existing members.
- **Charter and Guiding Principles:** Discussion on establishing a charter and guiding principles for operational and technical aspects of AI in the next meeting.
- **Group Visibility:** Decision to modify group.io settings to change WG visibility to non-public to ensure privacy.

No further comment from group. Meeting adjourned.